SMART INDIA HACKATHON 2026

Team ABNORMALIES

- **Problem Statement ID –** SIH26181
- **Problem Statement Title-** *AI Powered Personal Health Companion*
- **Theme-** MedTech/BioTech/HealthTech
- **PS Category-** *Hardware*
- **Team ID-**
- **Team Name-** *Abnormalies*

KIKI HEALTH COMPANION

Voice-First, Edge AI for Disaster-Resilient Health Monitoring

OUR IDEA

A health companion you talk to. Kiki sits on your desk, learns your routine, remembers you, and answers out loud in about a second — English or Hindi. It watches your vitals and the air and heat around you, and speaks up before things get bad. The model doing the talking runs on our own hardware.

HOW IT ADDRESSES THE PROBLEM

- Continuously monitors health and environmental signals.
- Detects abnormal patterns and emerging health risks early.
- Provides personalized warnings before risks escalate into emergencies.
- Supports users during heat waves, floods, pollution events and other disasters.

PROPOSED SOLUTION

BUILT FROM THE VERY SCRATCH BY US

INNOVATION/UNIQUENESS

- A companion, with sub-second spoken replies, memory of past conversations, and a personality. No smartwatch does this..
- Combines personal health data with real-time environmental risk context.
- Early-warning approach shifts healthcare from reactive response to preventive action.

Team Abnormalies

ABNORMALIES

SMART INDIA HACKATHON 2026

TECHNICAL APPROACH

TECHNOLOGIES

Hardware:

Desktop — Raspberry Pi 5, Hailo-8 accelerator, USB webcam, 28BYJ-48 stepper, LCD + OLED, IR sensors, bluetooth speaker.

Wearable — Waveshare ESP32-S3-Touch-AMOLED-1.75, MAX30102 (HR/SpO_2), QMI8658 IMU, 1000 mAh battery.

Inference — RTX 4060 laptop running llama.cpp (gemma-4-26B-A4B), whisper.cpp, OmniVoice.

Software:

Python 3.12 desktop code + WebSocket gateway • C++ firmware on ESP-IDF 5.5 • Next.js/TypeScript dashboard • MQTT • Tailscale • NumPy, pandas, scikit-learn, TensorFlow/TFLite, Arduino (PlatformIO), TFLite Micro

Cloud Services:

Cerebras (multi-step agents) • Gemini (background reasoning) • Open-Meteo API (weather + AQI)

METHODOLOGY

SENSOR

Collects real-time environmental and biometric data using MAX30102, QMI8658 IMU, and BMP280 sensors combined with Open-Meteo updates.

DETECT (ON DEVICE)

Processes data directly on the device to detect falls, classify physical activities, monitor heart rate and SpO2 levels, analyze visual context using OpenAI CLIP, and track CPCB AQI indices.

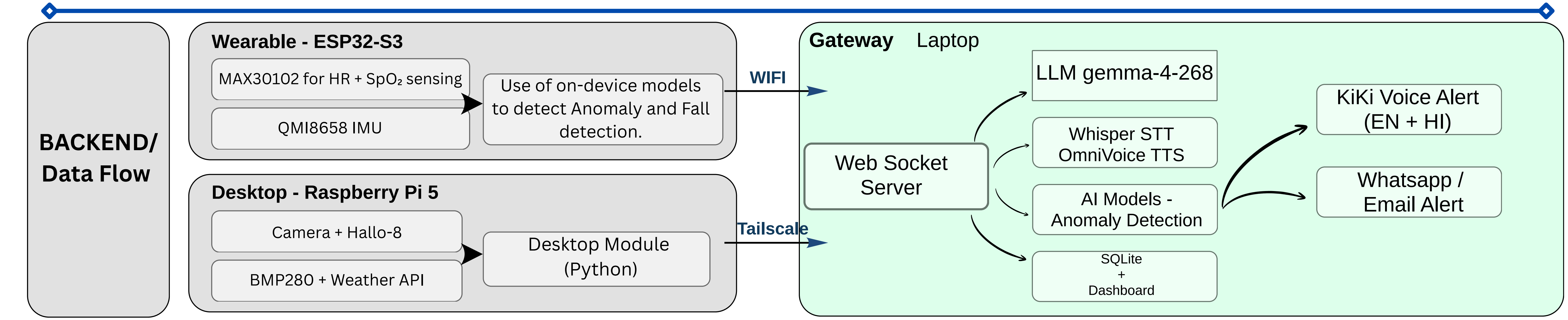
REASON (GATEWAY)

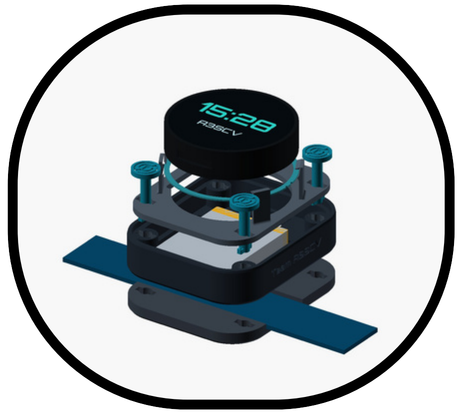
Executes high-level reasoning on the gateway via LLM, STT, and TTS pipeline with a warm slot instance and leverages

RESPOND

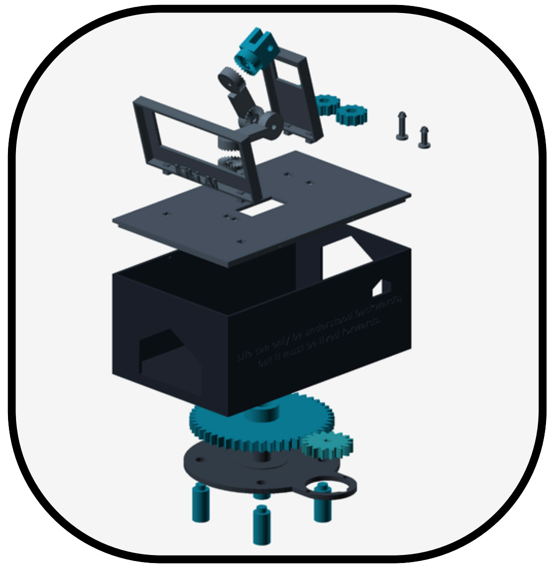
Delivers interactive feedback to the user using the Kiki voice system in English and Hindi, visual notifications on the AMOLED display, and external alerts via WhatsApp or email.

SYSTEM ARCHITECTURE





Wearable CAD Model



Desktop Kiki CAD Model

USER FACING SURFACES

- **KIKI Voice** — primary • elderly • hands-free
- **Wearable AMOLED** — on-body vitals + fall prompt
- **Desktop LCD + OLED** — room status • ambient
- **Web Dashboard** — caregiver • trends
- **Mobile Dashboard** — remote monitoring
- **WhatsApp / Email** — family emergency alert

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

- **Multi-source monitoring:** Wearable sensors for heart rate, SpO_2 and body temperature.
- **Integrated system:** Wearable, camera (chatbot) and mobile dashboard work together.
- **Real-time alerts:** Detects abnormalities and provides personalized guidance.
- **Prototype-ready:** Uses available hardware and software components.

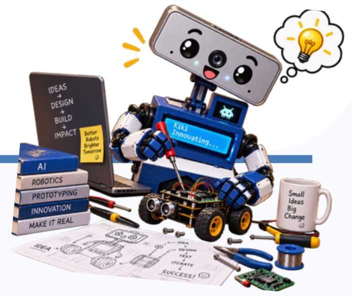
CHALLENGES & RISKS

01 **Sensor accuracy and calibration:** Heart rate and SpO_2 readings may vary with movement, placement and skin contact.

02 **Body temperature and location data issues:** Temperature readings can be affected by sensor limitations and environmental conditions; location data may be inconsistent.

MITIGATION STRATEGIES

- **Sensor calibration and signal filtering:** Calibrate during setup and apply filtering to reduce movement-related noise.
- **Warm prompt cache**: After every reply the whole conversation is prefilled back, so a follow-up costs ~10 tokens.
- **Lightweight edge models:** Optimize and compress (quants,mtp) models for faster, real-time inference.
- **Multi-reading confirmation:** Use consecutive readings and threshold checks before triggering critical alerts.



IMPLEMENTATION VIABILITY

- **Modular architecture:** Sensors, AI models and app can be developed independently and integrated incrementally.
- **Low infrastructure dependency:** On-device processing enables offline use and reduces reliance on continuous internet.
- **Rapid prototyping:** Built using existing, cost-effective hardware and open-source tools.
- **Scalable for future expansion:** Can support additional sensors, risk models and emergency scenarios.

IMPACT AND BENEFITS

IMPACT

ECOSYSTEM

Ensures Strict User Data Privacy: 100% on-device processing for vitals and fall detection eliminates raw data transmission and prevents cloud breaches.

Builds India's Edge-AI Health-Tech Capability: Fosters domestic innovation with an offline-first architecture, replacing cloud-dependent, imported wearables.

Advances Last-Mile Healthcare Connectivity: Aligns with **Ayushman Bharat Digital Mission** and **eSanjeevani** mandates bridging the digital infrastructure gap for underserved and aging populations.

BENEFICIARIES

Elderly Living Alone: Offers fall detection, spoken check-ins, and on-demand vitals (HR/SpO2) requiring absolutely zero smartphone or app literacy.

Outdoor Workers: Delivers preventative voice alerts based on heat-index and AQI, stopping heat stress before it becomes an emergency.

Chronic Patients: Automates the daily tracking of vitals and sleep trends without the need for frequent clinic visits or screen-based apps.

Caregivers & First Responders: Creates a closed-loop safety protocol (Fall → Voice Check → Automated WhatsApp/Email Alert) to ensure immediate action even when away.

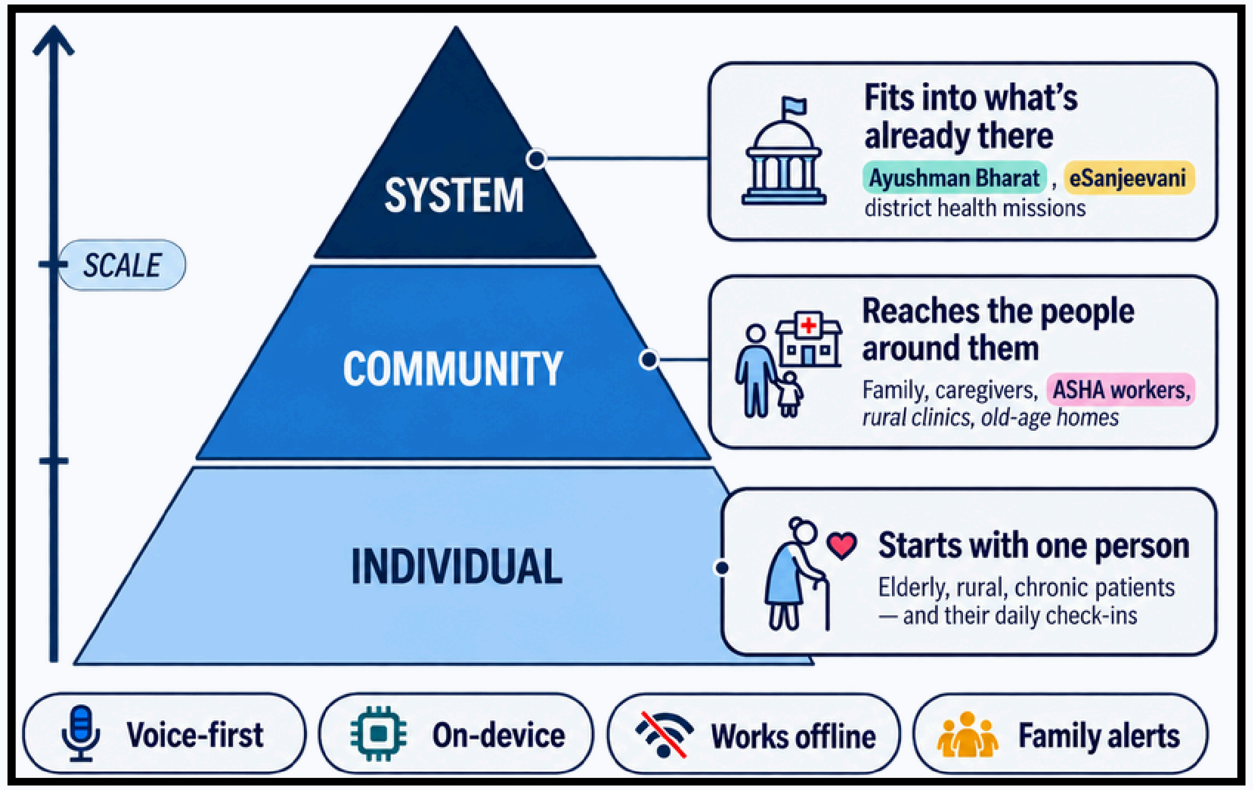
BENEFITS

Zero recurring cloud subscription for core safety features: Fall detection , vitals sensing , voice pipeline (llm+stt+tts) run on-device; cloud LLMs are used only for optional background reasoning.

Built on commodity, open hardware: Raspberry Pi 5, ESP32-S3, MAX30102, QMI8658 and Hailo-8, reducing dependence on proprietary hardware ecosystems.

Addresses a growing market: India's remote-patient-monitoring and wearable-health ecosystem provides a strong opportunity for scalable deployment.

Open, modular architecture: documented gateway + firmware separation enables pilots and deployments without vendor lock-in.



RESEARCH AND REFERENCES

RESEARCH PAPERS

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https://ieeexplore.ieee.org/abstract/document/11634798?casa_token=E-bVAo407bkAAAAA:t9DII2oUyjWUh1gqiu152vTVfQZ6x2F4bXFz3kNAqzL1VCYVzyG6BXIMwig6Fdk4gcFhd5LtEw

2) Usage of MAX30102 for SpO2 and HR monitoring-
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3) Raspberry Pi integration with MAX30102, BMP280 and MLX90614 and its efficiency-
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4) Fall Detection Feature-
https://www.tandfonline.com/doi/full/10.1080/07421222.2021.1990617#abstract

5) Multi Token Prediction (For Fast Responses)-
https://arxiv.org/abs/2404.19737

6) Health Insights-
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7) Edge Computing efficiency in Healthcare domain-
https://www.mdpi.com/1999-5903/16/9/329

REFERENCES

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https://github.com/ggml-org/llama.cpp

2)Raspberry Pi tutorials and docs-
https://pimylifeup.com/

3)AI Models-
A) https://huggingface.co/unsloth/gemma-4-26B-A4B-it-GGUF
B) https://huggingface.co/k2-fsa/OmniVoice
C) https://github.com/ggml-org/whisper.cpp

4)Waveshare Module and MAX 30102 -
https://docs.waveshare.com/ESP32-S3-Touch-AMOLED-1.75C

GITHUB REPOSITORY LINK

https://github.com/Abnormalies-SIH-Kiki/SIH-26181-Kiki-Health-Companion/tree/main

YOUTUBE VIDEO LINK

https://www.youtube.com/watch?v=SXq7fLjbRYM